



Experimental design and science writing

IB 431 – Behavioral Ecology
Fall 2025
September 23rd 2025

Announcements

- Brainstorming for proposal due this **Thursday, September 25th at 11:59pm**
 - Full description of assignment on Canvas
 - Generally – aim of this assignment is to show that you have reflected on feedback you got from peers/instructors and are beginning to think about the next steps for where your proposal will go, what experimental design will be, etc.

Outline

- I. Experimental design
- II. Science writing tips
- III. Continue proposal brainstorming (if needed)

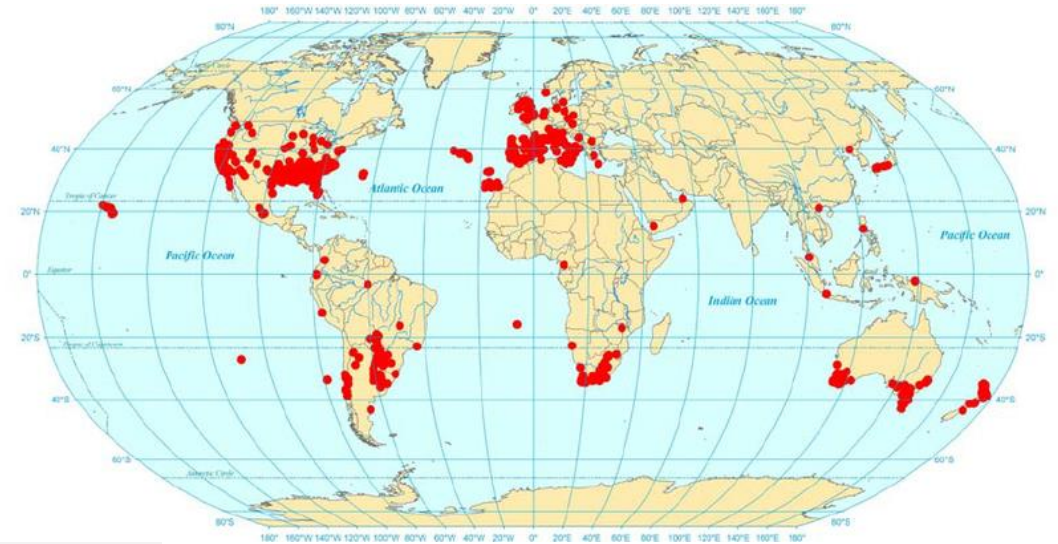
Quantifying behavior

- What is the behavior(s) being measured?
 - Needs to be operationalized and clearly defined
 - Distinguishable from other behaviors
 - Behavior is repeatable

Quantifying behavior

- What is the behavior(s) being measured?
- How will this behavior be quantified (i.e., how do you turn behavior into numbers)?
 - Frequency
 - Duration
 - Latency
 - Intensity

Example: aggressive behavior in Argentine ants



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- What is the behavior(s) being measured = interspecific aggression

Example: aggressive behavior in Argentine ants

- What is the behavior(s) being measured = interspecific aggression
 - **Ethograms** can help provide specific definitions of behaviors

Table 1

Ethogram of *L. humile* aggressive behaviors in the presence of a tethered *T. sessile* worker

Behavior	Description of behavior
Antennation	Contacting or feeling with antennae
Approach	Walking up to opponent
Prolonged contact	Close contact for extended period of time (>10 s); not fighting
Opening of mandibles	Spreading open mandibles towards opponent
Mounting	Climbing on top of opponent
Gaster tilt	Lifting gaster
Mandible lock	Using mandibles to grab opponent's mandibles
Biting	Biting into opponent
Fighting	Engaging in a fight with opponent, including close contact/attacks on opponent
Limb/body part detachment	Pulling off the limbs or body parts of opponent
Prolonged fighting	Fighting for extended period of time (>60 s)
Killing	Killing opponent during a fight

Example: aggressive behavior in Argentine ants

- How will this behavior be quantified:
 - Frequency = number of aggressive behaviors over 5 minutes
 - Duration = time spent engaging in aggressive behaviors
 - Latency = time it takes until first aggressive behavior observed
 - Intensity = how extreme is aggression (i.e., biting vs limb detachment)



Take a few minutes and consider these questions in context of your proposal and discuss with your table.

- What is the behavior(s) being measured?
- How will this behavior be quantified (i.e., how do you turn behavior into numbers)?
 - Frequency
 - Duration
 - Latency
 - Intensity
 - Something else?

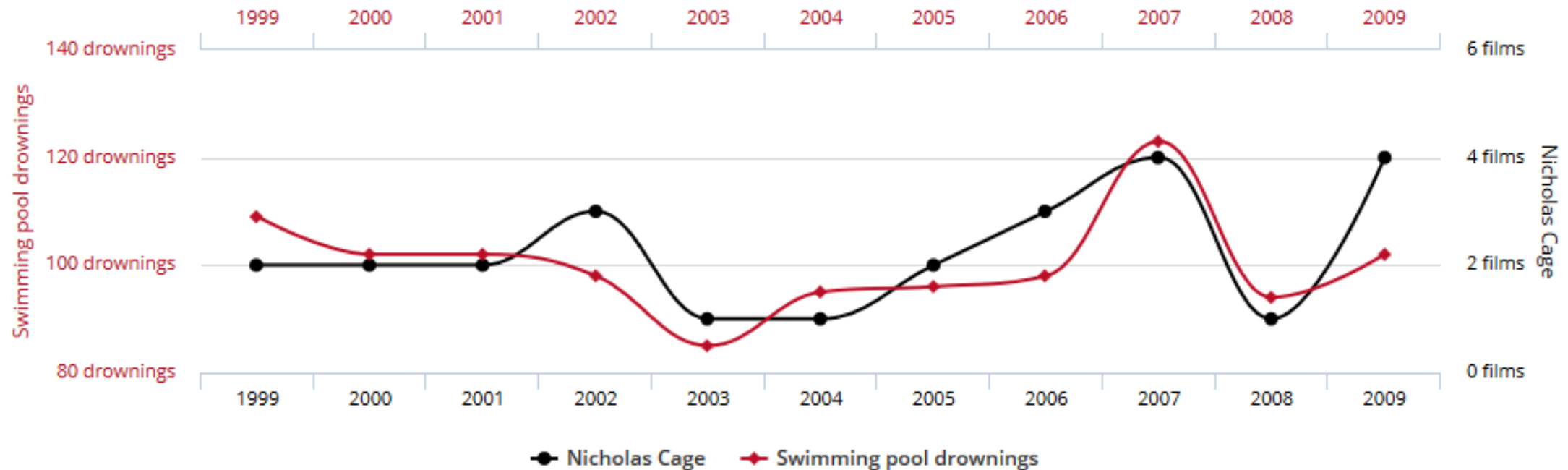
Observational studies vs. experiments

- **Observational study**
 - Observe patterns – no manipulation of variables or treatments
 - Can still have variables be measured but not controlled
 - Able to observe behavior undisturbed
 - But, limits our ability to make conclusions about causation (can only report correlations among variables)

Correlation $\neq$ causation

Number of people who drowned by falling into a pool
correlates with
Films Nicolas Cage appeared in

Correlation: 66.6% ($r=0.666004$, $p>0.05$)



When should you conduct observational studies?

- Logistical barriers to a controlled experiment
 - Whale migration patterns
 - Collective hunting behavior in wolves
- Ethical concerns
 - Predation rates on endangered species
- Behavior of interest is rare or does not occur in predictable manner
- Tracking evolution of behavior (especially for longer-lived species)

Observational studies vs. experiments

- **Experiment**
 - Controlled study where animals are exposed to specific treatments determined by the researcher
 - Can still do field experiment but need to be very careful with controlling for variables
 - Allows you to make conclusions about causation
 - Allows for replication of study under same conditions
 - Can control for confounding variables and thus be more confident in observed patterns

Independent and dependent variables

- Independent variable:
variable that you manipulate (e.g., temperature, habitat, social condition, age, environment, etc.)
 - Treatment vs. control groups
- Dependent variable:
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- Example: What is the effect of predation risk on activity in squirrels?
 - **Can you have multiple independent variables?**

Factorial design

- Assess the effects of multiple independent variables on your dependent variable by looking at interaction between independent variables
- Example: What is the effect of predation risk and food availability on activity in squirrels?
 - Predation: high vs. low
 - Food: high vs. low

Factorial design

Predation risk

Food
availability

High predation risk,
High food availability

Low predation risk,
High food availability

High predation risk,
Low food availability

Low predation risk,
Low food availability

Factorial design

- What if you wanted to look at more than two factors?
 - Three factors: $3 \times 3 = 9$ treatments
 - Four factors: $4 \times 4 = 16$ treatments, etc.
- Simpler, more concise studies are often the most effective

Limiting bias

- Observation bias =
 - Blinded observation (don't know treatment until after observing)
 - Multiple observers score behavior
 - Video recordings and automated analysis of behavior
- Seasonal bias = repeating study across different seasons
- Handling bias = give animal time to habituate to conditions before

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Academic science writing

- Discuss with your table:
 - What is your experience with academic science writing? Have you written a proposal, research paper, lab report, etc.? How do you feel about it? What challenges or obstacles have you encountered? Any questions or concerns that we can address?

Academic science writing

- Hourglass structure
 - Introduction = introduce general concepts then introduce system and specific questions/hypotheses
 - Discussion = start with specific interpretations and insight from results (or expected results) and then go into broader applications or significance



Possible structure for introduction

- P1: Introduce general concept
 - Last sentence – transition into general research gap
- P2: Discuss general research gap
- P3: Introduce your system
- P4: Discuss research gap from above in context of your system
- P5: Provide overview of study and state questions / hypotheses

Example: What is the effect of predation risk on activity in squirrels?

- P1: Understanding the impacts of predation risk on phenotype is important and can help us better understand the selective forces shaping prey behavior...
- P2: However, our understanding of the impacts of predation risk on prey activity remains limited...
- P3: Squirrels are a strong model system for exploring this...
- P4: Specifically, it is unclear if food availability has an impact on squirrel response to predation risk...
- P5: Here, I provided squirrels with different diets and explored their response to auditory cues of predators...

Possible structure for conclusion

- For research paper – discussion section
- For proposal – significance / broader impacts section
- Opposite of introduction – in introduction you narrowed down the topic
 - Start with specific implications / conclusions relevant to your system
 - Then expand scope and explain the impact of the results (or predicted results) on the same broader concepts you brought up in the first paragraph of the introduction

What if you don't know where to start?

- Start by establishing experimental design
- List off key concepts that emerge from your questions
- Write topic sentences for each of your paragraphs
- Identify sources for each of the paragraphs
- Type up bullet points for each paragraph
- Then work on writing it up in sentence form

Other random writing notes

- Avoid definitive language (always, all, never, etc.)
- Instead:
 - “Research suggests that...”
 - “There is evidence for...”
 - “X typically leads to an increase in Y”
 - “Data indicates X may occur...”
- Document with useful science writing phrases on Canvas (I will keep updating this – add your own ideas if you want)

Other random writing notes

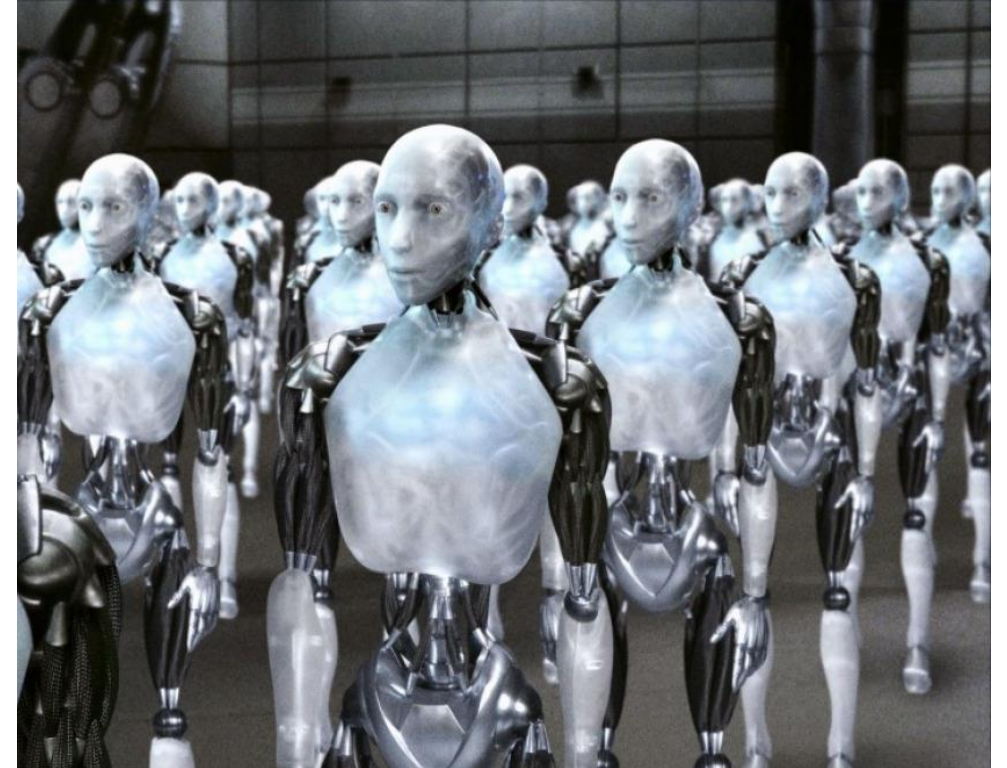
- Generally concise is better
 - Use references to your advantage – point reader to references if they want details instead of going into them yourself
 - Can also do this for methods – cite paper that uses similar method to yours

Other random writing notes

- Past tense for discussing published results; future tense for any proposed work
- Use of “we” and “I” is generally okay (maybe not other fields)
- Active voice (“We measured the aggression of ants”) over passive voice (“The aggression of ants was measured”)

Other random writing notes

- I would rather read your imperfect writing than read AI writing
- Goal of this class is to have your writing and scientific thinking develop throughout the semester – human writing with human feedback



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